COMS 3003A
Test 2 – Deferred version

7 May, 2024

This is a closed book test.
Time allowance: 60 minutes

Caution: if, in any of your answers, you define a Turing machine and claim that it solves a particular problem, you have to prove your claim.

1. **(20 points) Prove or disprove the following statements (i.e., if you believe the statement to be true, prove it; otherwise, provide a counterexample):**

   (a) **[redacted] points) If languages L_1 and L_2 are undecidable, then $L_1 \cup L_2$ is undecidable.**
   [redacted] is false. Let L_1 be any undecidable language. Then, $\bar{L_1}$ is also undecidable, but $L_1 \cup \bar{L_1}$ is [redacted]able, since $L_1 \cup \bar{L_1} = \Sigma^*$, and so it is decided by a Turing machine that accepts every inpu[redacted]

   (b) **(5 poi[redacted] Every countably infinite language is undecidable.**
   This is fa[redacted] The countably infinite language Σ^* is clearly decidable: it is decided by a Turing machine th[redacted]ccepts every input.

   (c) **(5 points) E[redacted] undecidable language is countably infinite.**
   This is true. E[redacted] language is countable. So, we need to show that every undecidable language is infinite[redacted]his is equivalent to every finite language being decidable. The latter is clearly true: if a lang[redacted]e L is finite, it can be decided by a Turing machine that first writes out all the words in L [redacted] then checks if the input word is among them.

   (d) **(5 points) If languages[redacted] and L_2 are undecidable, then $L_1 \cap L_2$ is undecidable.**
   This is false. Let L_1 be any [redacted]cidable language. Then, $\bar{L_1}$ is also undecidable, but $L_1 \cap \bar{L_1}$ is decidable, since $L_1 \cap \bar{L_1} =$ [redacted]nd so it is decided by a Turing machine that rejects every input.

2. **(10 points) Prove that the problem [redacted]determining whether the language of a Turing machine contains exactly 4 words is al[redacted]ithmically unsolvable, i.e., that the language $\{\langle M \rangle : |L(M)| = 4\}$ is undecidable. You [redacted] assume that the following problems are unsolvable:**

   - **non-self-acceptance for Turing machine[redacted]**

- **self-acceptance for Turing machines;**
- **acceptance for Turing machines.**

Suppose, for the sake of [redacted]tradiction, that this problem is solvable. Then, there exists a Turing machine, say M_4, that sol[redacted]t. Then, we can define a Turing machine M_A that decides acceptance:

(1) M_A, on input $\langle M, w \rangle$, [redacted]es down (the program of) a Turing machine $M_{M,w}$ that accepts as input a binary word x [redacted] works as follows:

  (a) $M_{M,w}$ simulates M [redacted];

  (b) If $M(w) = 1$, then [redacted], accepts x if x is either 00 or 01 or 10 or 11; otherwise $M_{M,w}$ rejects x.

(2) M_A calls M_4 with the argu[redacted]t $\langle M_{M,w} \rangle$.

(3) If $M_4(\langle M_{M,w} \rangle) = 1$, then M_A [redacted]cepts; $M_4(\langle M_{M,w} \rangle) = 0$, then M_A rejects.

Let's show that M_A, indeed, decide[redacted]ceptance:

- If $M_A(\langle M, w \rangle) = 1$, then $M_4($[redacted]$\rangle) = 1$, which means, by specification of M_4, that $|L(M_{M,w})| = 4$. This means that [redacted]$) \neq \varnothing$, which happens only if $M(w) = 1$.
- If $M_A(\langle M, w \rangle) = 0$, then $M_4(\langle M$[redacted]$) = 0$, which means, by specification of M_4, that $|L(M_{M,w})| \neq 4$. By definition of M[redacted] this happens only if $L(M_{M,w}) = \varnothing$, which means that $M(w) = 0$ or $M(w) = \infty$, i.e., th[redacted]$(w) \neq 1$.

Since acceptance is unsolvable, however, M_A [redacted] not exists, which gives us a contradiction.

3. **(20 points) Does there exist a bijection b[redacted]een the set of all deciders and the set of all undecidable decision problems? Pro[redacted]our statement. Your proof may not assume any fact proven in class or in textb[redacted]—if you need to use a fact in your proof, you need to prove it first.**

No, such a bijection does not exist. To show this, we s[redacted] that the set of deciders is countable, but the set of undecidable decision problems is uncounta[redacted]

Since every Turing machine, and hence every decider, c[redacted]e encoded as a string in a finite alphabet, the set of deciders is countable.

We can show using diagonilization that the set of all decision problems is uncountable. The set of all decidable problems is countable, since the cardinality of this set cannot exceed the cardinality of the set of all deciders. If the set of undecidable problems was also countable, then we could combine two enumerations to obtain an enumeration of the set of all decision problems, which is impossible, since this set is uncountable.

4. **(10 points) The first-order language of set theory contains a binary predicate symbol $\in$. Translate the following statements into this language (you are free to define any other symbols in terms of the ones listed above):**

  (a) **(5 points) There exists a set that contains exactly one element.**

(b) **(5 points) For every pair of sets X and Y, there exists a set whose elements are the common elements of X and Y.**

$$\forall x \forall y \exists z \forall u \, (u \in z \leftrightarrow u \in x \wedge u \in y).$$